

Extra credit 5. DigitalOcean Droplet

Not Saturday. A cheap permanent public endpoint.

New trigger. Same exits. Same harness. The graph does not change.

Cheapest Basic plan. Ubuntu 24.04 LTS. One vCPU. 1 GB RAM. About six dollars a month.

No new Python. You put extra credit 1 on a box.

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What you will build

The receiver is `solutions/extra_credit/s_ext_1_webhook`. nginx is the only public door. systemd runs uvicorn on loopback.

PIECE	PATH
<code>.env.example</code>	<code>labs/extra-credit/ext_5_digitalocean/.env.example</code>
<code>systemd unit</code>	<code>deploy/agent-webhook.service</code>
<code>nginx</code>	<code>deploy/nginx.conf</code>
<code>sol1 config</code>	<code>deploy/write-sol1-config.sh</code>
<code>handoff</code>	<code>deploy/call-sol1-enhancer.sh T001</code>
<code>smoke</code>	<code>deploy/smoke.sh</code>

Why bind to localhost

The agent holds a token. A default bind of `0.0.0.0` puts it on the public internet.

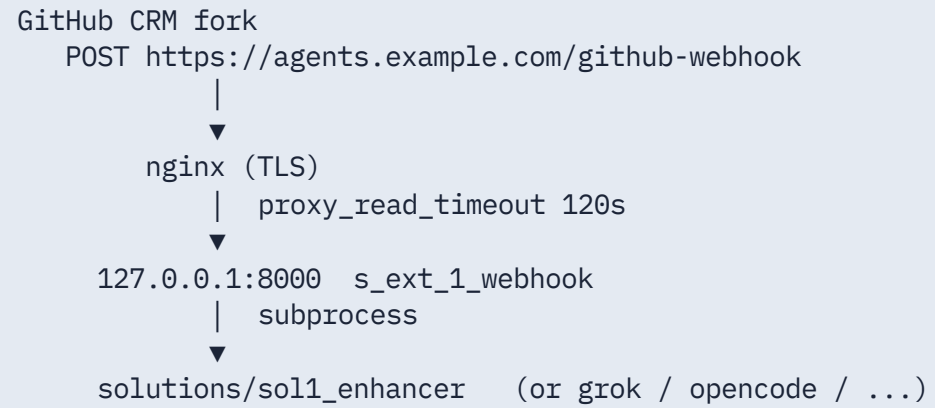
```
Internet —TLS→ nginx :443 —proxy→ 127.0.0.1:8000 unicorn
```

```
ProtectSystem=strict.NoNewPrivileges=yes.User agent.
```

Learning objectives

- Bootstrap Ubuntu 24.04 with one script
- Fill `/opt/agents/.env` from the example
- Terminate TLS with certbot
- Point the **CRM fork** webhook at `https://your-domain/github-webhook`
- Smoke-test HMAC locally before GitHub
- Swap `AGENT_BACKEND` without rewriting the unit

Starting architecture



One command on the box

```
ssh root@YOUR_DROPLET_IP
git clone https://github.com/YOUR-FORK/reliable-agentic-lab.git /opt/agents
bash /opt/agents/labs/extra-credit/ext_5_digitalocean/deploy/bootstrap.sh
```

Then edit `/opt/agents/.env`. Point DNS. `certbot --nginx -d your-domain`.

What bootstrap installs

- python3, venv, nginx, certbot, git, jq, task
- system user agent
- clone or pull /opt/agents
- venv at /opt/agent-env with requirements.txt
- .env from the example if missing (chmod 0600)
- optional clone of the CRM fork into work/northwind-field-crm
- write-sol1-config.sh then install.sh

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`.env` that matters

```
GITHUB_TOKEN=  
GITHUB_REPO=your-github-username/northwind-field-crm  
GITHUB_WEBHOOK_SECRET=  
AGENT_BACKEND=claude  
AGENT_MAX_ATTEMPTS=3  
AGENT_TIMEOUT=900  
WEBHOOK_HOST=127.0.0.1  
WEBHOOK_PORT=8000  
ANTHROPIC_API_KEY=  
WEBHOOK_DOMAIN=agents.example.com
```

`GITHUB_REPO` is the CRM fork, **not** the lab repo.

AGENT_BACKEND picks the folder

```
claude      -> solutions/sol1_enhancer
grok        -> solutions/sol1_enhancer_grok_build
opencode    -> solutions/sol1_enhancer_opencode
codex       -> solutions/sol1_enhancer_codex
agent-sdk   -> solutions/sol1_enhancer_agent_sdk
deep-agents -> solutions/sol1_enhancer_deep_agents
```

Grok is a good fit on a Droplet (you can `task trust` once). It is a poor fit on hosted Actions.

systemd unit. Loopback only.

```
ExecStart=/opt/agent-env/bin/python -m uvicorn \
    solutions.extra_credit.s_ext_1_webhook.webhook:app \
    --host 127.0.0.1 --port 8000
User=agent
EnvironmentFile=/opt/agents/.env
ReadWritePaths=/opt/agents
```

Working directory `/opt/agents`. `PYTHONPATH=/opt/agents`.

nginx. Two locations. Everything else 404.

```
location /health          { proxy_pass http://127.0.0.1:8000/health; }
location /github-webhook {
    proxy_pass http://127.0.0.1:8000/github-webhook;
    proxy_read_timeout 120s;
}
location / { return 404; }
```

`client_max_body_size 1m.` Replace `YOUR_DOMAIN` before certbot.

GitHub webhook

On the CRM fork (`GITHUB_REPO`):

- **Payload URL:** `https://your-domain/github-webhook`
- **Content type:** `application/json`
- **Secret:** same as `GITHUB_WEBHOOK_SECRET`
- **Events:** Issues, Issue comments, Check suites

Issue titles must include `[T001]`.

Smoke without GitHub

```
source /opt/agents/.env
bash labs/extra-credit/ext_5_digitalocean/deploy/smoke.sh
sudo -u agent bash -lc 'source /opt/agents/.env && \
/opt/agents/labs/extra-credit/ext_5_digitalocean/deploy/call-sol1-enhancer.sh T001 --print'
```

`smoke.sh` GETs `/health`, then POSTs a signed `issues.opened` body at loopback.

Expected live result

A 202 in the GitHub webhook log. A record in:

```
/opt/agents/solutions/extra_credit/s_ext_1_webhook/work/last-webhook.json
```

A task run in `solutions/sol1_enhancer` for that ticket.

```
journalctl -u agent-webhook -f
```

Troubleshooting

SYMPTOM	CAUSE	FIX
Droplet 502	uvicorn down, or proxy_pass wrong	<code>systemctl status agent-webhook</code>
401	secret mismatch	same string in GitHub and <code>.env</code>
503	secret empty	fill <code>.env</code> , restart unit
Clone failed	<code>CRM_OWNER</code> still the placeholder	<code>edit .env</code> , run <code>write-sol1-config.sh</code>
Grok skill not found	untrusted git root	<code>task trust</code> as the agent user

Recap

Permanent HTTPS. Same receiver as extra credit 1. Same exits as Lab 1.

The lid can close. The box keeps listening.

Next scale step: Fargate plus a queue. Same HMAC. Same 202. Worker is a second service.